INSTRUCTOR GUIDE & ANSWER KEY

SPHERE: MISSION CONTROL

INSTRUCTOR MANUAL, SAFETY GUIDANCE AND ASSESSMENT KEY

Scope: The activity is intended for introductory undergraduate engineering or physics teaching. Instructors should adapt the risk assessment, supervision ratio and assessment weighting to local institutional requirements.

1. LEARNING OUTCOMES

On completion, students should be able to:

- Distinguish conduction, convection and radiation in the test system.
- Apply Newton's law of cooling,

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt},$$

  and interpret the fitted cooling constant k (min^{-1}).
- Acquire and plot temperature data (T_{obs}) at defined time intervals ($t = 0 \text{ to } 15 \text{ min}$).
- Calculate residuals ($|\Delta T| = |T_{\text{obs}} - T_{\text{model}}|$) and MAE ($\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\Delta T_i|$), then discuss uncertainty and limitations of the lumped-parameter model.

2. 60-MINUTE LESSON ROADMAP

TIMELINE	LESSON PHASE	WHAT YOU & YOUR STUDENTS ARE DOING
$0\text{–}10\ \mathrm{min}$	1. Theory and concept check	Review the three heat-transfer modes, the water-bath boundary condition ($T_{\mathrm{env}} = 0.0^{\circ}\mathrm{C}$) and the limits of the spaceflight analogy. Complete the preliminary concept check.
$10\text{–}20\ \mathrm{min}$	2. Capsule Assembly	Assign teams their insulation setup (A: bare, B: bubble wrap, C: reflective film, D: multilayer assembly). Demonstrate probe placement and the pre-immersion leak test.
$20\text{–}35\ \mathrm{min}$	3. Fifteen-minute test	Add $100\ \mathrm{mL}$ of water at $80.0^{\circ}\mathrm{C}$ to each capsule. Measure the bath temperature, immerse the capsules, start the timer and record temperatures every $60\ \mathrm{seconds}$.
$35\text{–}48\ \mathrm{min}$	4. Analysis	Plot measured (T_{obs}) and predicted (T_{model}) temperatures, calculate residuals and MAE ($\mathrm{MAE} = \frac{1}{n}\sum \lvert \Delta T_i \rvert$), and compare the error with instrument accuracy ($\pm 0.5^{\circ}\mathrm{C}$).
$48\text{–}60\ \mathrm{min}$	5. Technical discussion	Compare the four configurations and discuss material mechanisms, model assumptions, uncertainty and design trade-offs.

3. LAB SETUP & SAFETY CHECKLIST

Complete and document the following controls before the session:

- **Hot-water control:** Do not use boiling water. Prepare water at $80.0^{\circ}\mathrm{C}$ and use heat-resistant gloves, eye protection and a stable pouring area in accordance with the local risk assessment.
- **Bath condition:** Prepare an ice-water slurry and measure its temperature. Do not assume an exact $0.0^{\circ}\mathrm{C}$ boundary without verifying.
- **Leak Prevention:** Check that students seal the probe opening using the approved lab method and complete a leak test before immersion.
- **Probe Placement:** Check that each probe hangs in the center of the liquid and isn't pressed against the plastic jar walls.

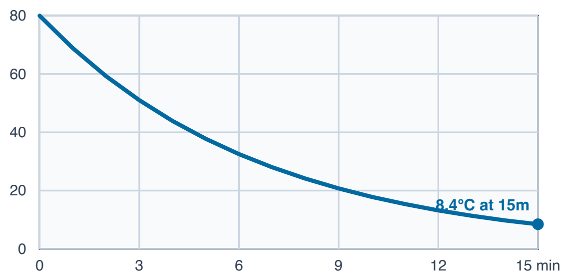
4. REFERENCE-MODEL OUTPUT

The table lists outputs from the assumed coefficients at $t = 15\ \mathrm{min}$, with $T_0 = 80.0^{\circ}\mathrm{C}$ and $T_{\mathrm{env}} = 0.0^{\circ}\mathrm{C}$. These are model values (T_{model}), not acceptance criteria for measured results.

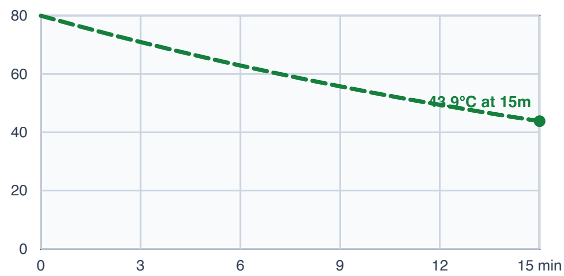
INSULATION TYPE	ASSUMED COOLING CONSTANT (k, min^{-1})	MODEL TEMP AT 15 MIN	QUALITATIVE INTERPRETATION
A. Bare Control	$k \approx 0.150\ \mathrm{min}^{-1}$	$\approx 8.4°\mathrm{C}$	Rapid heat loss through unshielded conduction into water and internal liquid circulation.
B. Bubble Wrap	$k \approx 0.040\ \mathrm{min}^{-1}$	$\approx 43.9°\mathrm{C}$	Sealed air cells reduce fluid motion and slow heat transfer.
C. Mylar Foil	$k \approx 0.121\ \mathrm{min}^{-1}$	$\approx 13.0°\mathrm{C}$	Reflective film is highly sensitive to conductive contact and leakage in the water-bath setup.
D. Multilayer assembly	$k \approx 0.015\ \mathrm{min}^{-1}$	$\approx 63.9°\mathrm{C}$	The combined layers increase effective thermal resistance through trapped air, reduced fluid motion and lower radiative exchange.

INDIVIDUAL PREDICTION CURVES ($T_0 = 80.0°\mathrm{C}$, $T_{\mathrm{env}} = 0.0°\mathrm{C}$)

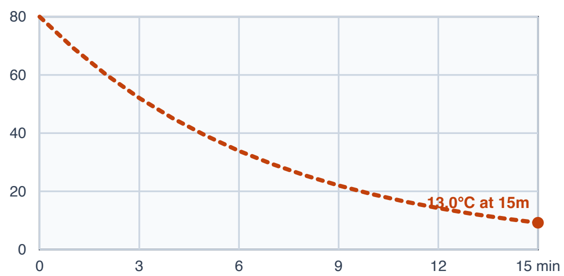
A. BARE CONTROL ($k \approx 0.150\ \mathrm{min}^{-1}$)



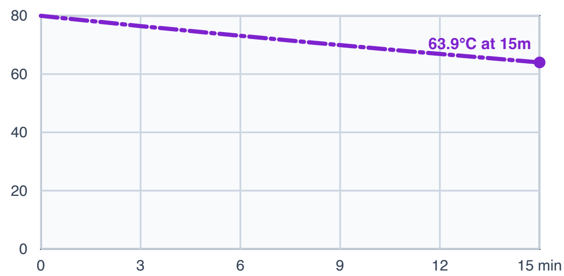
B. BUBBLE WRAP ($k \approx 0.040\ \mathrm{min}^{-1}$)



C. MYLAR FOIL ($k \approx 0.121\ \mathrm{min}^{-1}$)



D. MULTILAYER ASSEMBLY ($k \approx 0.015\ \mathrm{min}^{-1}$)



Custom mode: Mission Control also provides an adjustable k value for inquiry work. It is not shown as a fixed reference curve because its prediction changes with the instructor's selected value. **Optional extension:** A cotton-only configuration may be added as a fifth physical trial when time permits.

5. OFFICIAL SOLUTIONS & TEACHER ANSWER KEY

This key covers the eight-question extended reflection set in the student manual. The standalone task sheet contains six condensed practical inquiries. The separate 10-item online knowledge check auto-scores nine multiple-choice items; its Question 4 requires instructor review and is excluded from the automatic percentage. **Online key:** Q1 B, Q2 B, Q3 A, Q5 B, Q6 A, Q7 A, Q8 A, Q9 B, Q10 A.

Question 1: Why can reflective film provide limited insulation in direct water contact?

Sample answer: A reflective film can reduce radiative exchange, but it has little thickness and therefore provides limited resistance to conduction when it is in direct contact with the capsule and surrounding water. A low-conductivity spacer introduces trapped air and additional thermal resistance.

Marking guidance: award credit for distinguishing radiative performance from conductive resistance.

Question 2: What does the cooling constant (k) tell us?

Sample answer: Within the same test geometry and boundary conditions, a smaller fitted k means a slower exponential approach to T_{env} ($T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$). The value is an effective system parameter with units of min^{-1}; it is not the material thermal conductivity λ.

Question 3: Why can the multilayer assembly reduce cooling?

Sample answer: The materials introduce complementary resistance mechanisms: **Cotton** traps air and reduces effective conduction; **Bubble wrap** limits air movement within sealed cells; **Reflective film** reduces radiative exchange when an air gap is present. Interfaces between layers also add contact resistance ($k \approx 0.015\ \text{min}^{-1}$).

Question 4: Where could experimental errors come from?

Sample answer (Accept any two): (1) Sensor tip touching cold jar wall instead of water center. (2) Water leaking into jar through incompletely sealed lid or opening. (3) Ice bath warming up above $0.0°\text{C}$ during 15 minutes. (4) Water inside jar not stirred, causing thermal stratification.

Question 5: What happens if your ice bath warms up?

Sample answer: Students should measure the real bath temperature (e.g. $T_{\text{env}} = 4.5°\text{C}$) and type that new number into the T_{env} box in the Mission Control simulator to match actual classroom conditions ($T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$).

Grading Tip: Award credit if they mention updating T_{env} in the software.

Question 6: What if we double the water volume?

Sample answer: Doubling water ($100\ \text{mL} \to 200\ \text{mL}$) doubles mass of hot liquid ($Q = m \cdot c \cdot \Delta T$), holding twice as much thermal energy. With more heat to lose through roughly equal surface area, it cools **slower**, and cooling constant k **decreases**.

Grading Tip: Look for: cools slower and k decreases.

Question 7: Spacesuit Design Trade-Offs

Sample answer: Adding infinite layers makes the spacesuit heavier, bulkier, and stiffer, reducing joint mobility. Astronaut metabolic heat must also be rejected to prevent overheating.

Grading Tip: Accept joint flexibility/mobility, suit weight/bulk, or body heat buildup.

Question 8: How can the MAE be reduced?

Sample answer: (1) Use consistent lid-sealing method & leak-test capsule. (2) Position probe consistently near centre of liquid away from walls. (3) Swirl jar gently before reading. (4) Take readings at consistent recorded times ($\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |T_{\mathrm{obs},i} - T_{\mathrm{model},i}|$).

6. GRADING RUBRIC & CHECKLIST

CRITERIA	EXCELLENT (90-100%)	GOOD (75-89%)	NEEDS WORK (<75%)
Data Logging	All 15 minutes filled with accurate readings, absolute differences, and helpful notes.	Completed with only 1-2 missed minutes; minor calculation mistakes.	Multiple blank rows; missing calculations.
Graphing	Clean, accurate points plotted with a smooth line; dashed theoretical curve clearly shown.	Points plotted with minor curve scaling inaccuracies.	Points not connected; axes mislabeled or missing lines.
Analysis Questions	Clear, thoughtful answers explaining conduction, convection, radiation, and MLI synergy (Q1-Q8).	Good understanding of heat transfer with minor details missing.	Confuses conduction with radiation; incomplete answers.